



IonoMoni: ionospheric monitoring based on dual-frequency data from single GNSS station

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Abstract

The ionosphere has a significant impact on Global Navigation Satellite System (GNSS) signals, introducing refraction and diffraction effects that can severely degrade positioning performance. At the current peak of Solar Cycle 25, escalating ionospheric space weather events and disturbances have further intensified adverse effects on GNSS applications. On the other hand, dual-frequency GNSS observations enable effective monitoring of ionospheric conditions, serving as a crucial foundation for the mitigation of ionospheric effects on GNSS positioning and the advancement of research into the physical mechanisms of ionospheric variability. In this context, we developed a software tool named IonoMoni in C++ for calculating ionospheric monitoring parameters. IonoMoni enables processing of observation data from GPS, BDS, Galileo, and GLONASS to compute the Rate of TEC Index (ROTI), Along Arc TEC Rate (AATR), and to extract both Slant Total Electron Content (STEC) and Vertical Total Electron Content (VTEC) using the Carrier-to-Code Leveling (CCL) method and the Undifferenced and Uncombined Precise Point Positioning (UCPPP) method. Extensive batch processing and visualization scripts are provided for the time series and distribution of STEC, VTEC, ROTI and AATR. Experimental validation of ionospheric monitoring parameters was conducted using GNSS observations of IGS stations at different regions. The results demonstrate that IonoMoni reliably produces precise TEC and ionospheric perturbation products, making the software well suited for ionospheric monitoring while providing a solid foundation for GNSS-based ionospheric research.

Keywords GNSS · STEC · VTEC · Ionospheric perturbation · ROTI · AATR

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Introduction

The ionosphere is a highly variable propagation medium located in the upper atmosphere of the Earth, primarily formed by the interaction of high-energy electromagnetic radiation and charged particles from the Sun with atmospheric constituents (Schunk and Nagy 2009; Brekke 2013). The main impact of the ionosphere on Global Navigation Satellite System (GNSS) signals can be classified into two categories: (1) phase advance and group delay, and (2) ionospheric scintillation (Dubey et al. 2006; Linty et al. 2018). Phase advance and group delay arise from ionospheric refraction. The ionospheric ray-bending effect introduces sub-millimeter-level (less than ± 0.3 mm) positioning errors, which are negligible compared with the first-order ionospheric effect (Petrie et al. 2010). Owing to the frequency-dependent refractive index of the ionospheric plasma, the group velocity of GNSS signals decreases while the phase

velocity increases, leading to ionospheric range errors that may reach several tens of meters (Mainul and Jakowski 2012). Ionospheric scintillation originates from diffraction and scattering of transionospheric radio signals by plasma density irregularities, producing stochastic fluctuations in GNSS signal amplitude and carrier phase, known as amplitude and phase scintillation, respectively (Kintner et al. 2007). Scintillation also exhibits a clear latitude dependence and occurs more frequently at low and high latitudes: phase scintillation is generally more frequent and stronger at high latitudes, whereas amplitude scintillation is more commonly observed at low latitudes (Hernández-Pajares et al. 2011; Vasylyev et al. 2022). In response to ranging error and ionospheric scintillation, researchers have proposed a variety of key ionospheric parameters.

STEC is a fundamental parameter in ionospheric research, representing the integrated electron density along the propagation path from a GNSS satellite to the receiver, including both ground-based receivers and low Earth orbit (LEO) radio occultation (RO) receivers. STEC can typically be extracted from GNSS code and carrier-phase observations using Carrier-to-Code Leveling (CCL) method or Undifferenced and Uncombined Precise Point Positioning method (Ren et al. 2021). The ranging error induced by ionospheric phase advance and group delay is proportional to STEC, and STEC can be mapped to VTEC using a mapping function. Ionospheric maps constructed on the basis of VTEC can be used to correct the associated ranging errors, which is particularly important for improving the positioning accuracy of single-frequency users. In addition, ionospheric maps can provide key ionospheric constraints that facilitate rapid convergence of Precise Point Positioning (PPP) (Hernández-Pajares et al. 2009; Jakowski et al. 2011; Liu et al. 2021b; Jerez et al. 2022).

Although ionospheric delay can be modeled and corrected, ionospheric scintillation is difficult to model because of its stochastic nature and intermittently occurring behavior. Scintillation is commonly monitored and warned using scintillation indices. S_4 and σ_ϕ are widely used indices for characterizing amplitude and phase scintillation, respectively, and are directly provided by dedicated ionospheric scintillation monitoring receivers (ISMRs). However, ISMRs typically operate at sampling rates of 50–100 Hz, which imposes stringent requirements on onboard storage and data transmission bandwidth and is often associated with higher equipment cost. These factors limit large-scale global deployment of ISMRs. In contrast, geodetic GNSS receivers are widely deployed worldwide, and scintillation indices derived from geodetic GNSS observations such as the Rate of TEC Index (ROTI) and the Along-Arc TEC Rate (AATR) make it feasible to observe ionospheric scintillation

over large areas or even on a global scale (Luo et al. 2020; Liu et al. 2022; Li et al. 2022).

ROTI was proposed by Pi et al. (1997) and is defined as the standard deviation of the rate of TEC within a given time interval. ROTI can be computed from low-rate GNSS observations (e.g., 30 s sampling) to monitor ionospheric phase scintillation and to detect ionospheric irregularities including equatorial plasma bubbles (EPBs) and polar cap patches (Krakowski et al. 2006; Nishioka et al. 2008). In recent years, multiple studies have used ROTI to investigate ionospheric responses to events such as volcanic eruptions and earthquakes (Saito 2022; Vesnin et al. 2023), and ROTI has also been increasingly incorporated into high-precision GNSS positioning algorithms, including PPP, PPP-RTK (Precise Point Positioning Real-Time Kinematic, PPP-RTK) and network RTK (NRTK), to optimize stochastic models and cycle-slip detection thresholds (Luo et al. 2022; Li et al. 2025a; Zhang et al. 2025). AATR is also a scintillation-sensitive index proposed by Sanz et al. (2014), computed as the root mean square (RMS) of the along-arc TEC rate for all observable satellites over a one-hour interval. By contrast, ROTI is computed separately for each satellite and is typically computed every five minutes or over even shorter intervals. Consequently, AATR provides broader spatial sensitivity than a single satellite track while requiring a substantially smaller data volume than ROTI. Achieving a comparable monitoring scale with ROTI typically requires observations from multiple receivers to generate ROTI maps (Jacobsen and Andalsvik 2016; Juan et al. 2018).

To support operational high-precision GNSS applications and ionospheric scintillation monitoring, the aforementioned parameters are routinely generated as standardized global ionospheric products by several analysis centers. Within the International GNSS Service (IGS) framework, the combined Global Ionospheric Maps (GIMs) are generally regarded as the most accurate reference solution and are obtained as a weighted average of the GIMs provided by seven Ionosphere Associate Analysis Centers (IAACs), such as the Center for Orbit Determination in Europe (CODE), the Jet Propulsion Laboratory (JPL), and the Universitat Politècnica de Catalunya (UPC) (Roma-Dollase et al. 2018; Liu et al. 2021a; Martire et al. 2024). In terms of scintillation monitoring, global ROTI maps are provided by the IGS, the German Aerospace Center (DLR), and Nagoya University's Institute for Space-Earth Environmental Research (ISEE) (Cherniak et al. 2018; Anoruo et al. 2022; Kriegel et al. 2026). The European Space Agency (ESA) provides global AATR maps through its Space Weather Service Network (Paul et al. 2025). However, open-source software for computing the above ionospheric parameters directly from raw multi-GNSS observations is still relatively scarce. Therefore, this study developed IonoMoni, a dedicated

ionospheric parameter computation software. Built upon GREAT-PVT (Li et al. 2025b), an open-source software for PPP and RTK, IonoMoni supports the calculation of ROTI and AATR indices and the extraction of VTEC and STEC using CCL and UCPPP methods, based on multi-GNSS observation data for ionospheric monitoring and research.

Development environment

The IonoMoni software is developed based on the Microsoft Visual Studio 2022 integrated development environment, using the C++ programming language and following the C++17 standard. The project adopts CMake as the build system, which is primarily used to organize the source code structure and manage the linking of external libraries. In addition, IonoMoni uses Extensible Markup Language (XML) files for parameter configuration, which not only allows flexible control over operation modes, input paths, and functional modules, but also enables seamless integration with batch processing scripts for efficient automated workflows.

Functions of IonoMoni

The functions of IonoMoni include STEC extraction, VTEC conversion, and the calculation of ROTI and AATR indices. These functions are all based on dual-frequency GNSS observations. At present, IonoMoni supports the following constellations and signals: GPS (L1, L2, L5), GLONASS (G1, G2), Galileo (E1, E5a, E5b, E5, E6), and BDS (B1I, B2I, B3I). For each constellation, the user can flexibly select different dual-frequency signal combinations through the XML configuration file for the computations. IonoMoni supports processing 30-s sampling interval RINEX 2.x and 3.x GNSS observation files and accepts as inputs precise products from any analysis center that provides standard SP3 precise orbits, CLK precise clocks, and differential code bias (DCB) products in Bias-SINEX format. Furthermore, IonoMoni provides auxiliary scripts for batch processing and visualization functions. The processing workflows of the IonoMoni software are shown in Fig. 1. Considering the paper length, more comprehensive descriptions of the mathematical models will be provided in the IonoMoni user manual.

STEC extraction based on CCL method

The essence of CCL approach is to utilize pseudorange and carrier phase observations within each continuous arc to resolve STEC. In the processing workflow of the CCL_STEC mode in IonoMoni, the geometry-free linear

combinations, L4 and P4, are formed by the difference of dual-frequency carrier phase and pseudorange observations. Subsequently, the carrier-to-code leveling algorithm is applied within each continuous carrier phase arc (Zhang 2016). In addition, the CCL_STEC framework is generic and can be extended to LEO-based GNSS observations for STEC extraction.

STEC extraction based on UCPPP method

In comparison with the CCL method, UCPPP can fully incorporate external constraints including precise satellite orbits and clock products. GNSS observation data are pre-processed, and the Kalman filter is employed to jointly estimate all parameters. The estimated ionospheric parameter is converted to STEC for output (Zhang et al. 2012; Li et al. 2025b). Furthermore, IonoMoni applies DCB corrections to the STEC values extracted by both the CCL and UCPPP methods.

VTEC conversion from STEC and mapping function

In both the CCL_STEC and PPP_STEC workflows, the obtained STEC can be converted to VTEC by dividing STEC by the ionospheric mapping function. IonoMoni provides five mapping functions, including the Single Layer Model (SLM), the Modified Single Layer Model (MSLM), the F&K mapping function proposed by Foelsche and Kirchengast (2002), the Ou mapping function proposed by Ou (1996), and the Faselow mapping function proposed by Sovers and Faselow (1987). Users can select the mapping function through the XML configuration. The features and typical ionospheric heights of each mapping function model are presented in Table 1 (Sovers and Faselow 1987; Ou 1996; Yang et al. 2014; Yuan et al. 2021; Chen et al. 2025). In particular, the MSLM is derived from the SLM by tuning its parameters to best fit the extended slab model mapping function used by JPL, which leads to the specific effective ionospheric height of 506.7 km (Feltens et al. 2018; Wielgosz et al. 2021).

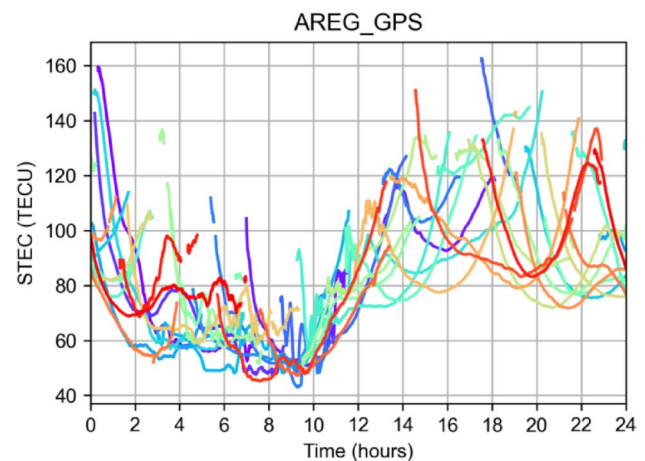
ROTI index based on GNSS dual-frequency data

The processing workflow of the ROTI module in IonoMoni follows the method proposed by Jacobsen (2014). The ROT is computed as the temporal difference of the geometry-free carrier phase combination L4 between successive epochs, and ROTI is calculated as the standard deviation of the ROT values within a sliding window. In previous studies of ionospheric scintillation based on 30 s sampling data, a 5-min window has been widely adopted (Pi et al. 1997; Basu et al. 1999; Sripathi et al. 2011); Jacobsen (2014) pointed out that

Table 1 Characteristics and ionospheric effective heights of various mapping function methods

Mapping function	Ionospheric height (ground users)	Ionospheric height (spaceborne users)	Features
SLM	350/450 km	–	(1) Simple formula, physically intuitive; (2) Error increases at low elevation angles.
MSLM	506.7 km	–	(1) Introduces empirical corrections based on SLM; (2) Higher accuracy than standard SLM, especially at low elevation angles.
F&K	450 km	$h_{orbit} + 450$ km (integral method) $h_{orbit} + 1100$ km (centroid method)	(1) Derived from spherical geometry; (2) Suitable for spaceborne users.
Ou	350/450 km	–	(1) Ionospheric projection function with segmented values adapted to elevation angle; (2) Suitable for low elevation angles.
Fanselow	Central height: 350 km; Lower boundary: 315 km (h-35); Upper boundary: 420 km (h+70); Thickness: 105 km	–	(1) Simulates the ionosphere as a shell with finite thickness, approximating the actual vertical profile than a single thin shell; (2) Effective at low elevation angles.

Both ROTI and AATR are scintillation indices derived from TEC variation information obtained from carrier-phase measurements. However, since AATR is an RMS quantity, it additionally accounts for the dispersion of the STEC rate compared to ROTI. As a result, AATR is highly sensitive to both the spatial gradients and temporal variations of STEC, and can not only characterize scintillation intensity but also reflect the average level of ionospheric perturbations over a relatively large region (Sanz et al. 2014). Large Scale Traveling Ionospheric Disturbances (LSTIDs) are large-scale TEC perturbations (wavelengths of hundreds to thousands of kilometres, periods from tens of minutes to hours) that produce STEC rate variations across many satellite-receiver links in a region. And the AATR index aggregates along-arc rates across multiple satellite-receiver links and computes an RMS naturally highlights region-scale perturbation associated with LSTIDs. For example, Ferreira et al. (2020)

**Fig. 2** STEC of different GPS satellites using CCL_STEC at station AREG on DOY 001, 2016

showed that the response of AATR to mid-latitude LSTIDs is clearly superior to that of ROTI, making AATR more suitable as a near-real-time indicator and warning metric for the occurrence of mid-latitude LSTIDs.

Batch processing and visualization capabilities

All functions in IonoMoni support processing data from multiple stations within a single day. Batch processing Python scripts are provided to facilitate automated analysis of multi-day, multi-station datasets. Additional Python tools enable time series visualization of STEC, VTEC, ROTI, ROT, and AATR results, as well as the generation of regional VTEC distribution maps.

Results

To validate the various functional modes of IonoMoni, we selected observation data from six IGS stations to experimentally validate the ROTI, AATR, and STEC results obtained using both the CCL and UCPPP methods.

CCL_STEC results

To validate the accuracy of the CCL_STEC module in IonoMoni software, GNSS data from the AREG station on Day of Year (DOY) 001 of 2016 were selected. As presented in Fig. 2, the extracted STEC is in excellent agreement with the results of Ren et al. (2021).

PPP_STEC results

To validate the accuracy of the PPP_STEC extraction module from IonoMoni software, three stations located at high

latitude (YAKT), mid-latitude (BJFS), low latitude (BAKO) were selected for STEC time series computation. The GNSS Analysis software for Multi-constellation and multi-frequency Precise positioning (GAMP) enables precise STEC extraction using undifferenced and uncombined PPP (Zhou et al. 2018). Figure 3 shows the STEC time series produced by IonoMoni and GAMP and the corresponding residuals. The residual time series indicates that the residuals gradually decrease as the PPP solution converges and become stable after convergence. The mean residuals are only 0.0118 TECU for YAKT, 0.0906 TECU for BJFS, and 0.1626 TECU for BAKO. Overall, the STEC results extracted by IonoMoni are highly consistent with the GAMP-derived results across stations at different latitudes.

VTEC results

The hourly VTEC regional distribution maps (UT 00:00–11:00) over Central Europe for DOY 100, 2023 are presented in Fig. 4.

VTEC for single station can be converted by using the ionospheric mapping function and STEC extracted from CCL or UCPPP methods. GNSS data located at Europe from 3 IGS stations were downloaded. And CCL method is applied to extract STEC from GNSS data. The VTEC at each station are obtained by projecting STEC with SLM mapping function.

ROTI results

To validate the accuracy of the ROTI index computation module in IonoMoni, we performed a comparison with M_IONO, an ionospheric scintillation index computation and

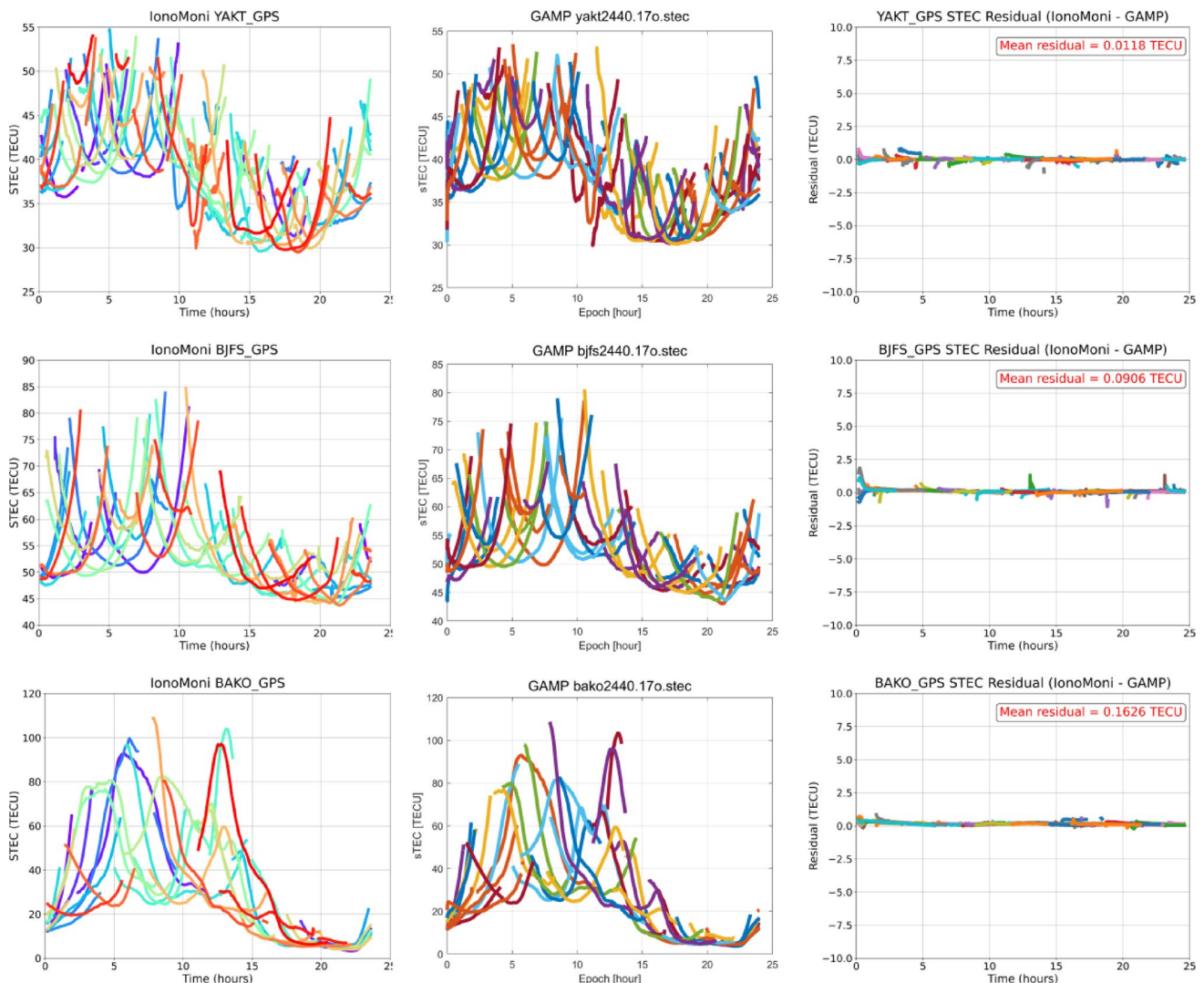


Fig. 3 Comparison of STEC at three stations (YAKT, BJFS, BAKO) using the UCPPP method by IonoMoni and GAMP on DOY 244, 2017

Fig. 4 Hourly VTEC regional distribution maps (UT 00:00–11:00) at Europe for DOY 100, 2023

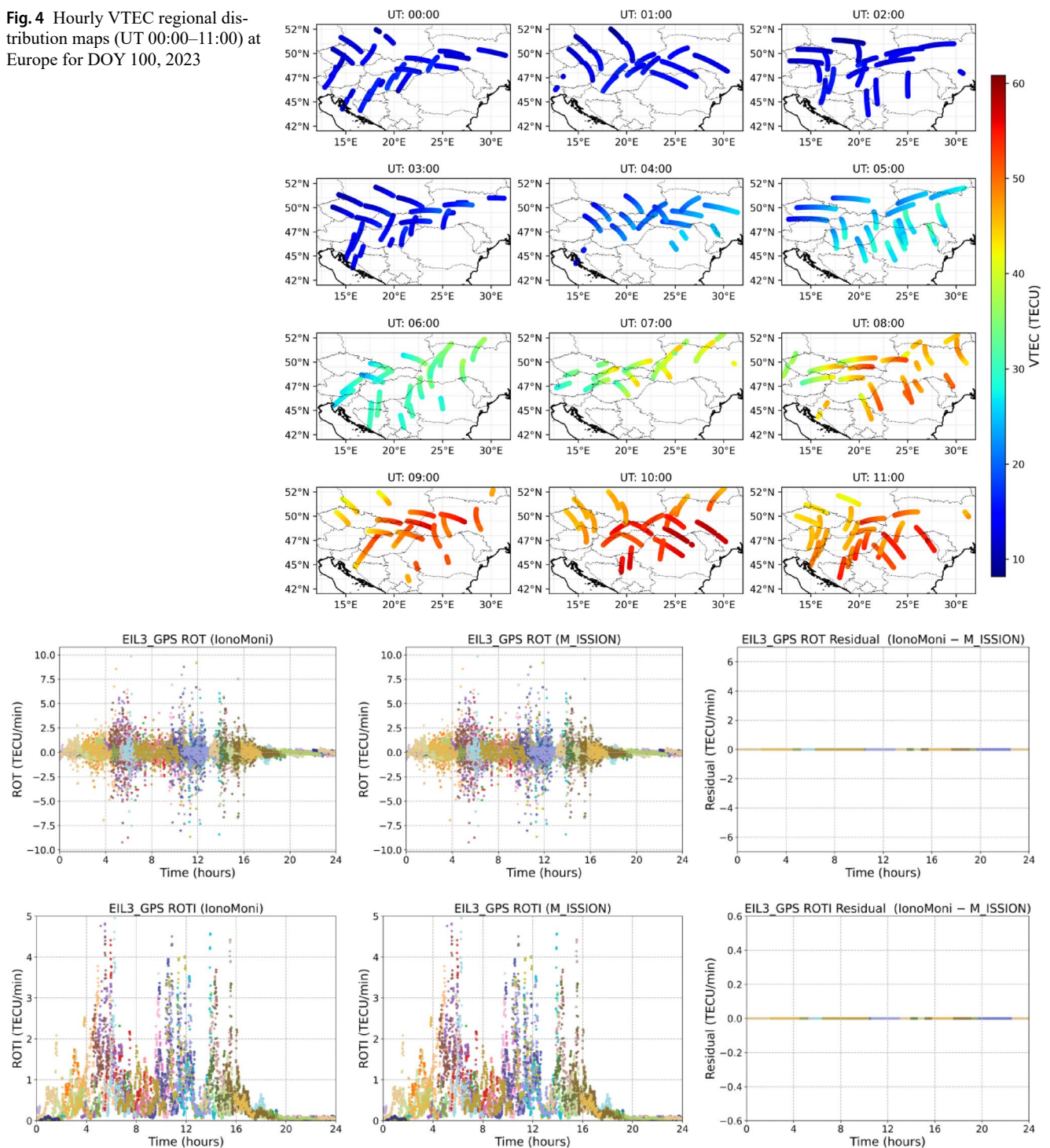
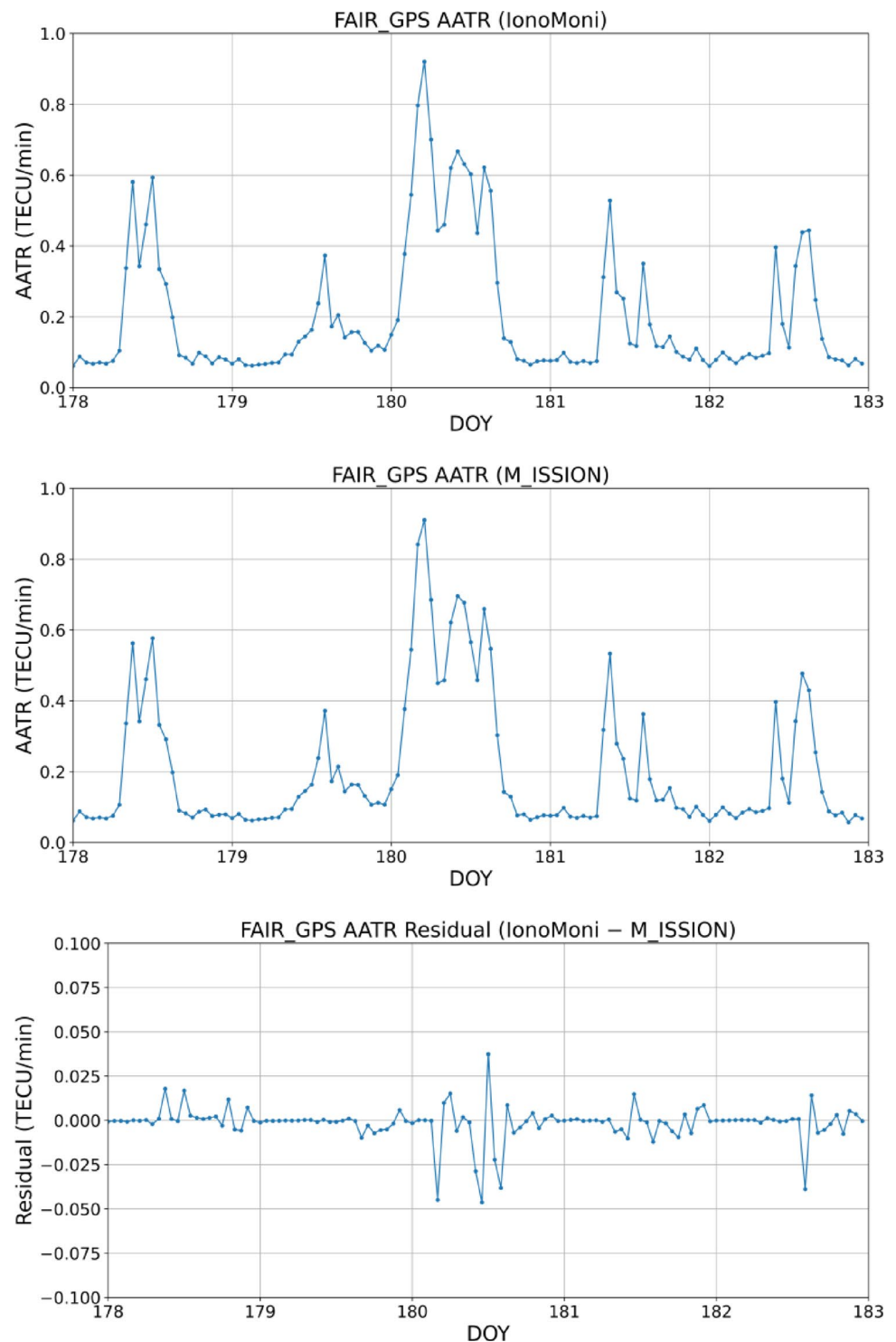


Fig. 5 Comparison of ROT and ROTI at EIL3 station by IonoMoni and M_ISSION on DOY 180, 2024

analysis software (Zhao et al. 2026). GPS observation data from station EIL3 on DOY 180 of 2024 were processed by IonoMoni and M_ISSION to derive ROT and ROTI, and Fig. 5 shows the time series results together with the corresponding residuals. The residual time series indicate that the ROT and ROTI time series produced by the two software

packages are identical. Under the same parameter configuration, IonoMoni consistently reproduces the M_ISSION-derived ROT and ROTI results, demonstrating the accuracy of the ROTI module implemented in IonoMoni.

Fig. 6 Comparison of AATR at FAIR station by IonoMoni and M_SSION on DOY 178 to 192, 2024



AATR results

To validate the accuracy of the AATR index computation module in the IonoMoni software, GPS observation data from the FAIR station on DOY 178 to 182 of 2024 were selected. Figure 6 shows the AATR time series produced by IonoMoni and M_SSION and the corresponding

residuals. The results show that the AATR estimates from IonoMoni are highly consistent with the AATR estimates from M_SSION in both magnitude and temporal variation. The residuals are positively correlated with the AATR level. The differences between the two AATR time series mainly arise from different implementations of cycle-slip detection and satellite-position interpolation, which lead to

slight discrepancies in both the values and the number of instantaneous AATR samples included in the hourly AATR computation. The resulting AATR differences are more than one order of magnitude smaller than the AATR values and can therefore be neglected in practice.

Conclusion

To support a wide range of ionospheric research, the open-source software IonoMoni was developed to enable efficient extraction, visualization, and batch processing of key parameters, including STEC, VTEC, ROTI, and AATR. The software supports simultaneous generation and export of results across multiple GNSS systems and is compatible with various GNSS observation data formats, significantly reducing data selection constraints. Reproducing experiments from previous studies has enabled comprehensive validation of IonoMoni core functional accuracy.

The ionospheric parameters obtained by IonoMoni demonstrate a high degree of consistency with both published literature and established reference software. Future updates will include the support for higher sampling rates and may incorporate real-time processing capabilities.

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Author contributions Yuzhi Zhang, Yihong Zhang, Qi Liu jointly participated in software development; Yuzhi Zhang and Qi Liu conducted the experimental research and wrote the manuscript; Manuel Hernandez-Pajares and Enric Monte-Moreno conducted the review and analysis; Heng Yang, Chenxiang Wang, Chuanding Zhang, Li Yang performed the formal analysis; Huiling Yu and Wenkai Meng were responsible for data curation and validation. All authors reviewed the manuscript.

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Data availability The IonoMoni software is available at <https://github.com/qiliu2025/IonoMoni>; All the data can be obtained from the IGS data center of Wuhan University (<ftp://igs.gnsswhu.cn/>) and earthdata (<https://www.earthdata.nasa.gov/centers/cddis-daac>).

Declarations

Competing interests The authors declare no competing interests.

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